

Nurses with tablets and bikers with smartphones join Ebola fight

By Joseph D'Urso

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LONDON (Thomson Reuters Foundation) - While health experts in laboratories globally have tried to devise ways to end the largest Ebola epidemic on record, technology experts have also been coming up with new ways to attack the virus - using smartphones and Ebola-proof tablets.

Recording and tracking the victims of Ebola in the West African nations of Sierra Leone, Guinea and Liberia has proved a major challenge since the outbreak started in December 2013 and spread rapidly, killing more than 11,000 people.

Ivan Gayton, an emergency coordinator with the medical charity Medecins Sans Frontieres (MSF), said inefficient tracking was not helping, with workers in treatment centers sharing information about patients by shouting it over a fence to colleagues in a low-risk zone to record by hand.

For workers were aware that it was not just bodily fluids that could spread the disease but also contaminated paper and clipboards which triggered the idea of an Ebola-free tablet.

So Gayton teamed up with a Dutch developer Pim De Witte, founder of a technology company called Whitespell, to devise an Ebola-proof tablet aided by funding from Google.

An off-the-shelf device is given a polycarbonate shell to make it waterproof and chlorine resistant and charged wirelessly so it does not need to be taken in and out of contamination zones.

Workers punch in data about patients, which can be stored, tracked and analyzed externally. The interface is optimized for people wearing steamy goggles and thick plastic gloves.

All the software, including the design for the shell which is made by 3D printing, is open source, meaning anyone can download and improve it. Data can be stored on the tablet when there is no signal, and uploaded later.

"It's actually really complicated to make it simple," Gayton told the Thomson Reuters Foundation.

BIKES IN TECHNOLOGY

He said the software took a while to roll out because of extensive testing but went into use in February when the virus was on the wane. This month Liberia was declared Ebola-free.

"We were late to the party ... we had to err on the side of first do no harm," said Gayton, but added that he hoped the device would be useful in future disease outbreaks.

Tracking victims around West Africa also proved difficult.

When health workers arrived in the Tonkolili district of northern Sierra Leone, they had limited information about the names, location and details of villages they needed to help.

"A lot of information was missing," said Laura Nic Lochlainn, a fellow at the RIVM National Institute for Public Health and the Environment in the Netherlands, who traveled to Sierra Leone this year with an MSF team to find a solution.

For a two week trial, researchers employed locals to scoot around the province on small motorbikes known as okadas, collecting household, health and population data from villages on simple smartphones.

They traveled in pairs, one riding the motorbike and one using a GPS-enabled smartphone running an Android operating system, preloaded with a specially designed, simple program for storing the necessary information.

When they arrived in a village they interviewed a village leader or representative to gather as much information as possible, and log GPS coordinates, essential in a region where village names are often duplicated or spelt differently.

Nic Lochlainn said it takes a long time to learn to use the sophisticated satellite devices usually used for mapping but users could master this software in hours and the data let experts assign Ebola cases to specific villages more accurately.

The scheme covered 950 villages in two weeks, and the cost was "very modest" compared with sending foreign aid workers into the field or commissioning detailed satellite imagery, she said.

"Technology can get in the way if it's complicated, but if it's simple it can only do good things," Nic Lochlainn told the Thomson Reuters Foundation.

Reporting By Joseph D'Urso, Editing by Belinda Goldsmith

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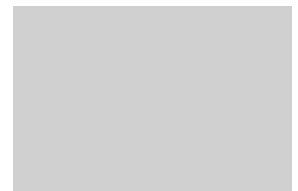
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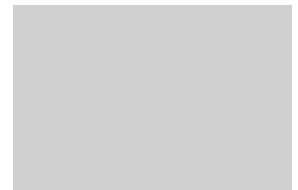
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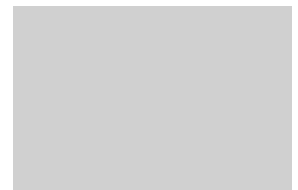
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